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Traffic state reconstruction in heterogeneous traffic flow from multi-source partial observations: A self-supervised and physics-informed framework☆

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ABSTRACT

Traffic state reconstruction in mixed traffic is fundamentally challenged by heterogeneous partial observability, where sparse mobile observations from connected and automated vehicles (CAVs) coexist with localized fixed observations from roadside units (RSUs). Existing methods often struggle to simultaneously address source heterogeneity, realistic continuous missing patterns, and the recovery of physically meaningful congestion propagation structures. To address these issues, this study proposes a source-aware physics-enhanced grid fusion framework for lane-level traffic state reconstruction. The method first projects heterogeneous CAV and RSU observations onto a unified time–space grid and encodes observed values, observability masks, and explicit spatiotemporal coordinates into a multi-channel representation. A block-wise masking self-supervised strategy is then introduced to simulate realistic continuous missing regions and enforce contextual reconstruction learning. In addition, a source-aware grid fusion network is developed to explicitly model mobile and fixed sensing sources and adaptively integrate them through gated fusion. Physics-guided regularization terms, including state-aware wave propagation, congestion continuity, and anisotropic smoothing constraints, are further incorporated to improve the structural plausibility of the reconstructed traffic state field. Experiments on the NGSIM US-101 dataset show that the proposed framework outperforms representative self-supervised baselines, achieves robust reconstruction under different CAV penetration rates, and better preserves the continuity and directional consistency of low-speed congestion bands in the time–space plane.

1. Introduction

In Intelligent Transportation Systems (ITS), the accurate perception and reconstruction of traffic states serve as a fundamental prerequisite for traffic monitoring, control optimization, and proactive traffic management [1–4]. Driven by advancements in the

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Internet of Vehicles (IoV), roadside sensing, and multi-source monitoring technologies, the observational data available within modern traffic systems has become increasingly abundant. However, challenges regarding the incompleteness, heterogeneity, and locality of traffic observations have become progressively prominent [5–7]. This is particularly evident in mixed traffic environments where Connected and Automated Vehicles (CAVs) and Human-Driven Vehicles (HVs) will coexist for the foreseeable future, rendering the system characteristically partially observable. Extant research indicates that the disparities in sensing capabilities, response mechanisms, and operational characteristics between different vehicle types significantly influence the formation, evolution, and observability of traffic states. Consequently, achieving reliable state estimation under limited CAV penetration rates has emerged as a critical challenge in mixed traffic research [8–12].

Within this context, CAVs equipped with communication and sensing capabilities can transmit real-time operational data, such as velocity and position, thereby functioning as mobile sensors distributed along vehicle trajectories. Nevertheless, the efficacy of such mobile observations is constrained by factors such as CAV penetration, spatio-temporal vehicle distribution, and sampling imbalances, often resulting in sparse, discrete, and irregular coverage. In contrast, roadside units (RSUs) provide continuous local observations within fixed spatial segments; however, their sensing range is typically localized, making it difficult to reconstruct large-scale, continuous traffic states independently. Therefore, under mixed traffic conditions, integrating mobile observations from CAVs with fixed observations from RSUs to achieve continuous traffic state reconstruction under heterogeneous, partially observable conditions has become a pivotal issue in contemporary traffic state estimation (TSE) research [13–16].

Research concerning traffic state estimation and reconstruction can be broadly categorized into three categories. The first category comprises model-driven approaches based on traffic flow theory and state-space models, including kinematic wave theory, the Cell Transmission Model (CTM), and Kalman filtering along with its various extensions. These methods typically possess strong physical interpretability and are capable, to a certain extent, of characterizing the spatio-temporal evolution of traffic states. Recent studies have further attempted to integrate traffic flow conservation laws, fundamental diagram (FD) relationships, or computational graph methods into state estimation frameworks to enhance reconstructibility and robustness under sparse observation conditions [17–19]. However, these approaches often rely on stringent a priori assumptions, such as regularized sensor layouts, fixed state-transition structures, or relatively idealized traffic evolution mechanisms. Consequently, their modeling flexibility and adaptability remain constrained when addressing multi-source heterogeneous inputs composed of CAV mobile observations and RSU fixed observations, especially in the presence of localized continuous data gaps and complex disturbance propagation inherent in mixed traffic environments [13,17–19].

The second category comprises data-driven traffic state estimation methods. In recent years, alongside the rapid advancement of deep learning in spatio-temporal modeling, architectures such as Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), Graph Neural Networks (GNNs), and Transformers have been widely deployed for traffic state recovery, imputation, and prediction tasks. These approaches exhibit significant advantages in learning nonlinear traffic evolution features and characterizing complex spatio-temporal dependencies, offering novel paradigms for reconstructing complete traffic states from incomplete observations. Concurrently, multi-source data fusion has emerged as a dominant trend within this field. For instance, the joint estimation of Electronic Toll Collection (ETC) data, fixed detectors, trajectory data, and connected vehicle data has been proven to effectively enhance both the coverage and accuracy of traffic state recovery [13,20–25]. However, a substantial portion of existing data-driven methods typically assumes relatively uniform observation sources or relies on simple concatenation for joint modeling. Such approaches seldom explicitly distinguish the intrinsic disparities between different observation sources regarding their sampling mechanisms, spatial coverage, information density, and reliability. Particularly in mixed traffic scenarios, the mobile sparse observations provided by CAVs and the localized continuous observations from RSUs exhibit markedly different spatio-temporal distribution characteristics. Without targeted modeling tailored to these heterogeneous sensing sources, models often struggle to fully exploit the complementary relationships between these two types of observations [14–16,22,23,26].

The third category of research focuses on continuous spatio-temporal traffic representation and implicit field learning. These methods represent traffic states as functions defined over a continuous spatio-temporal domain, achieving traffic field recovery, interpolation, or super-resolution reconstruction through the mapping of coordinates to state variables. By transcending the representation limits of traditional discrete grid-based modeling, these approaches allow models to perform queries and recovery at arbitrary spatio-temporal locations, thereby providing greater flexibility for traffic state reconstruction. Relevant studies indicate that continuous representation, Implicit Neural Representations (INRs), and direction-aware spatio-temporal modeling can enhance the generalization and representational capacity of traffic state reconstruction under sparse observation conditions [27,28]. Concurrently, some researchers have attempted to improve traffic field imputation by leveraging anisotropic kernels, image restoration paradigms, or generative imputation frameworks [29]. However, existing continuous field learning and missing-data recovery methods predominantly focus on fitting continuous state representations from uniform sparse observations, often failing to account for the heterogeneous sensing structures formed by the combination of CAVs and RSUs in real-world traffic monitoring. Furthermore, their optimization objectives typically prioritize point-wise fitting errors, offering limited attention to structural features such as propagation bands in the time-space plane, the continuity of congestion waves, and directional smoothness. Consequently, under complex partially observable conditions, these methods are prone to issues where numerical recovery appears plausible, yet the propagation structures remain blurred, congestion bands become fragmented, or the state fields suffer from over-smoothing [30,31].

In summary, current research into the reconstruction of traffic states under conditions of mixed traffic still faces several critical issues that require further refinement. First, there is a lack of a unified and source aware modeling framework capable of reconciling heterogeneous sensing sources, specifically mobile observations from CAVs and fixed observations from RSUs, which possess fundamentally different sampling mechanisms. Second, existing training paradigms rely heavily on full supervision or simulated random missingness, creating a significant discrepancy with the localized continuous missing patterns commonly encountered in

real world traffic sensing. Third, many methodologies prioritize the reconstruction accuracy of the state values themselves, while providing insufficient modeling of the underlying structural characteristics, such as propagation patterns in the temporal and spatial plane, the continuity of congestion bands, and directional organizational features. While extant studies on multi source fusion, physics informed learning, and continuous representation have provided valuable insights into these challenges, there remains substantial room for expansion regarding unified modeling specifically tailored to the heterogeneous and partially observable scenarios inherent in mixed traffic flows [13,18,19,22,27].

Addressing the aforementioned issues, this research develops a traffic state reconstruction method specifically for heterogeneous and partially observable environments within mixed traffic flows. This approach integrates multi source sensing fusion, self supervised learning, and physics informed guidance. Given that velocity directly characterizes traffic operational performance and sensitively reflects phenomena such as congestion wave propagation, queue expansion, and local bottlenecks, velocity is selected as the core representation variable for the traffic state. Specifically, the sparse mobile observations provided by CAVs and the localized fixed observations from RSUs are first unified into regularized temporal and spatial grids for individual lanes. This process simultaneously encodes observed values, observability masks, and explicit coordinates to construct a multi channel input representation suitable for full field reconstruction. Subsequently, to address the localized continuous missingness common in real world sensing, a block wise masking self supervised training mechanism is designed. By artificially constructing continuous temporal and spatial gaps within observed regions, the model is compelled to utilize contextual information to recover masked areas, thereby learning the continuous propagation structures of traffic states. At the architectural level, a source aware grid fusion network is introduced. This network utilizes a CAV branch, an RSU branch, and a gated fusion mechanism to achieve structured modeling and adaptive integration of heterogeneous observation sources. The reconstruction is completed through a U Net styled multi scale temporal and spatial encoder decoder backbone. Finally, to prevent the model from degrading into a simple numerical interpolator, this study incorporates state aware wave propagation constraints, congestion wave continuity constraints, and anisotropic smoothing constraints. These elements enhance the physical plausibility and organizational integrity of the results from the perspective of traffic wave propagation theory. Such designs align with contemporary research trends in physics guided state estimation and sparse observation recovery, while further extending toward heterogeneous sensing fusion scenarios under mixed traffic conditions.

Compared with existing work, the primary contributions of this study are as follows:

(1) **A unified single lane temporal and spatial grid representation** is constructed for mixed traffic environments where CAVs and HVs coexist. By incorporating observed values, observability status, and positional information into the input space, it provides a structured modeling foundation for reconstruction under multi source partial observations.
(2) **A block wise masking self supervised training mechanism** is proposed. This organizes the learning process to align with real world data missing patterns, enabling the model to recover continuous missing regions from local and global contexts rather than relying solely on full supervision.
(3) **A source aware grid fusion network** is designed. This achieves adaptive integration of heterogeneous sensing sources by explicitly distinguishing between CAV and RSU observation branches through a gated fusion approach.
(4) **Physics guided constraints** are introduced based on traffic propagation laws. By incorporating state aware wave propagation, congestion wave continuity, and anisotropic smoothing, the model recovers not only the numerical state values but also the underlying temporal and spatial propagation structures.

The remainder of this paper is organized as follows. Section 2 presents the methodology, primarily detailing the problem definition, the network architecture, the self supervised training objectives, and the physics guided regularization constraints. Section 3 describes the experimental design, the data processing procedures, and the evaluation metrics. Section 4 demonstrates the experimental results of the model in traffic state reconstruction tasks, followed by a detailed analysis from the perspectives of multi source fusion, self supervised masking, and physical constraints. Finally, Section 5 concludes the paper and discusses future research directions.

2. Methodology

To address the heterogeneous partial observability arising in mixed traffic flow, where connected and automated vehicles (CAVs) coexist with human-driven vehicles (HVs), this study proposes a source-aware physics-enhanced grid fusion framework for traffic state reconstruction in the time–space domain. The key idea is to integrate sparse mobile observations provided by CAVs and local fixed observations provided by roadside units (RSUs) into a unified structured representation, and then recover a continuous traffic state under partial observability through self-supervised learning and physics-guided regularization. As illustrated in Fig. 1, the proposed framework follows four main steps. First, heterogeneous observations from CAVs and RSUs are projected onto a regular lane-level time–space grid, where observed values, observation masks, and explicit spatiotemporal coordinates are jointly encoded to form a multi-channel input tensor. Second, a block-masking self-supervised strategy is introduced to simulate realistic continuous missing patterns in observed regions, thereby forcing the model to infer masked regions from both local and global context. Third, a source-aware grid fusion network is designed to explicitly model CAV-based mobile observations and RSU-based fixed observations in separate branches and adaptively integrate them through a gating mechanism within a U-Net-style encoder–decoder architecture. Finally, physics-guided regularization terms, including state-aware wave propagation, congestion continuity, and anisotropic smoothing constraints, are incorporated to improve the structural plausibility of the reconstructed traffic field.

Through this design, the proposed method provides a unified solution that simultaneously addresses heterogeneous sensing fusion, self-supervised reconstruction under partial observations, and physical consistency in traffic state propagation.

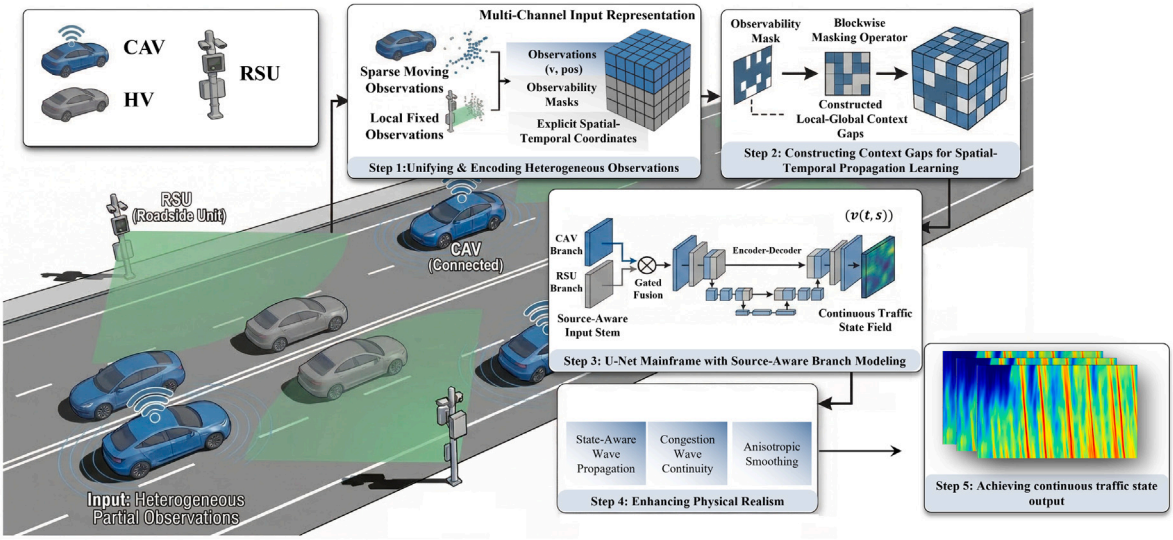


Fig. 1. Overall idea of heterogeneous partial-observation traffic state reconstruction in mixed traffic flow.

2.1. Problem definition

In real-world traffic sensing environments, traffic states are rarely fully observable. This issue becomes particularly prominent under mixed traffic flow conditions, where CAVs coexist with HVs, resulting in a traffic system with heterogeneous and incomplete observations. On the one hand, CAVs equipped with sensing and communication capabilities can continuously upload their own operating states, such as speed and position, thereby forming mobile observations distributed along vehicle trajectories. However, due to the limited penetration rate of CAVs, uneven vehicle distribution, and irregular spatiotemporal sampling, these observations are typically sparse, discrete, and highly nonuniform in the time–space domain. On the other hand, RSUs can provide relatively continuous measurements within fixed local road segments, but their sensing range is usually spatially limited and thus insufficient to independently support large-scale continuous traffic state recovery.

Motivated by this, this study focuses on traffic state reconstruction under heterogeneous partial observations in mixed traffic flow, namely, recovering a continuous and complete time–space traffic state representation from CAV-based mobile observations and RSU-based local fixed observations.

Let the ground-truth traffic state on lane l be denoted by

$$x_l(t,s), \qquad (t,s) \in \mathcal{T} \times \mathcal{S}, \tag{1}$$

where t denotes time, s denotes the longitudinal road position, and $\mathcal{T}$ and $\mathcal{S}$ denote the temporal and spatial domains of interest, respectively. Since speed can directly characterize traffic operating conditions and sensitively reflect typical traffic phenomena such as congestion wave propagation, queue spillback, and localized deceleration regions, speed is adopted in the current implementation as the core traffic state variable :

$$x_l(t,s) = v_l(t,s). \tag{2}$$

Although speed is adopted as the target state variable in this study, the proposed framework is not restricted to speed reconstruction. Other macroscopic variables, such as density and flow, can also be represented on the same time-space grid and reconstructed using a similar source-aware grid fusion structure. In trajectory-based datasets, these variables can be obtained by aggregating microscopic vehicle records within each grid cell. Nevertheless, speed is selected in the current implementation because it can be directly observed from vehicle trajectories and CAV records, and is less sensitive to vehicle-counting errors and boundary-crossing definitions than density or flow. In addition, speed is highly informative for characterizing congestion wave propagation, queue spillback, and localized deceleration regions. By contrast, density and flow require additional aggregation and may be more affected by grid resolution, sampling sparsity, and partial observability. Future work will extend the framework toward joint reconstruction of speed, density, and flow fields.

Accordingly, the objective is to learn a mapping from multi-source partial observations to a continuous traffic state field:

$$\hat{x}_l(t,s) = f_\theta\left(\mathcal{O}_c^{(l)}, \mathcal{O}_r^{(l)}\right), \tag{3}$$

where $\mathcal{O}_c^{(l)}$ and $\mathcal{O}_r^{(l)}$ denote the sets of CAV observations and RSU observations on lane l, respectively, θ denotes the model parameters, and $\hat{x}_l(t,s)$ is the reconstructed traffic state. Since the present study adopts speed as the state variable, the output can be further

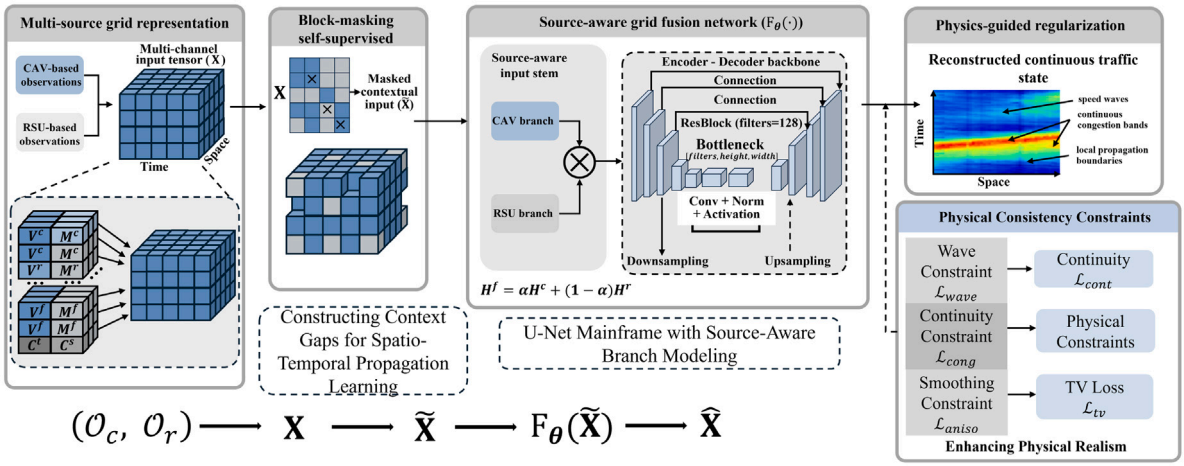


Fig. 2. Architecture of the proposed network.

written as

$$\hat{x}_l(t,s) = \hat{v}_l(t,s). \tag{4}$$

Rather than directly fitting irregular sparse observations in a pointwise manner, the proposed method first projects heterogeneous observations from different sensing sources onto a regular time–space grid and then performs full-field reconstruction in the grid domain. In this way, multi-source observations with substantially different sampling mechanisms, spatial coverage patterns, and information densities are transformed into a structured representation suitable for deep spatiotemporal learning.

2.2. Network architecture

The overall architecture of the proposed framework is illustrated in Fig. 2. To reconstruct a continuous traffic state field from heterogeneous partial observations, we develop a source-aware physics-enhanced grid fusion network. Instead of adopting a pointwise coordinate-querying paradigm as in implicit neural representations, the proposed framework directly reconstructs the entire traffic state field on a regular time–space grid. This design is more suitable for capturing the global structural characteristics of traffic states in the time–space plane, especially continuous congestion bands, speed waves, and local propagation boundaries.

Overall, the network consists of four major components: (1) a multi-source grid representation module that converts CAV-based and RSU-based observations into a unified multi-channel input tensor; (2) a block-masking self-supervised module that generates structured missing regions for contextual reconstruction learning; (3) a source-aware grid fusion network that explicitly models the two sensing sources and reconstructs the full traffic state field; and (4) a physics-guided regularization module that constrains the reconstructed field from the perspectives of propagation consistency and structural plausibility. The overall information flow can be summarized as

$$(\mathcal{O}_c, \mathcal{O}_r) \longrightarrow \mathbf{X} \longrightarrow \widetilde{\mathbf{X}} \longrightarrow F_\theta(\widetilde{\mathbf{X}}) \longrightarrow \hat{\mathbf{X}}, \tag{5}$$

where $\mathbf{X}$ denotes the original multi-channel observation tensor, $\widetilde{\mathbf{X}}$ denotes the masked contextual input after block masking, $F_\theta(\cdot)$ denotes the grid fusion network, and $\hat{\mathbf{X}}$ denotes the reconstructed continuous traffic state field. In the current implementation, $\hat{\mathbf{X}}$ corresponds to the reconstructed speed field.

2.2.1. Multi-source grid representation

Each lane is discretized into a regular time–space grid, where the temporal and spatial dimensions are divided into T and S grid cells, respectively. In the current implementation, a spatiotemporal resolution of 5 s × 5 m is adopted. Based on this discretization, the input tensor for lane l is defined as

$$\mathbf{X}^{(l)} \in \mathbb{R}^{C\times T\times S}, \tag{6}$$

Where, C denotes the number of input channels. In the current implementation, the input tensor for lane l consists of the following eight channels:

$$\mathbf{X}^{(l)} = \left[\mathbf{V}^c, \mathbf{M}^c, \mathbf{V}^r, \mathbf{M}^r, \mathbf{V}^f, \mathbf{M}^f, \mathbf{C}^t, \mathbf{C}^s\right]. \tag{7}$$

Specifically, $\mathbf{V}^c$, $\mathbf{V}^r$, and $\mathbf{V}^f$ denote the state observations associated with CAV sensing, RSU sensing, and fused sensing, respectively. In the current implementation, these observations correspond to speed measurements. $\mathbf{M}^c$, $\mathbf{M}^r$, and $\mathbf{M}^f$ are the corresponding binary observability masks, while $\mathbf{C}^t$ and $\mathbf{C}^s$ represent the normalized temporal and spatial coordinates, respectively.

For an arbitrary grid cell (t, s), if at least one CAV observation is available within the cell, we set $M^c(t, s) = 1$, and otherwise $M^c(t, s) = 0$. Similarly, if the cell falls within the sensing coverage of an RSU and contains a valid RSU observation, we set $M^r(t, s) = 1$, and otherwise $M^r(t, s) = 0$. As a result, $\mathbf{M}^c$ typically exhibits a sparse and irregular spatiotemporal sampling pattern distributed along vehicle trajectories, whereas $\mathbf{M}^r$ represents a locally continuous observation pattern over fixed spatial segments.

To provide the network with a coarse-grained reference for joint observations, we further construct a fused observation channel. The corresponding fused mask is defined as the element-wise union of the observable regions from the two sensing sources, namely

$$\mathbf{M}^f = \max(\mathbf{M}^c, \mathbf{M}^r). \tag{8}$$

Accordingly, the fused observation value $\mathbf{V}^f$ is defined as

$$V^f(t,s) = \begin{cases} V^c(t,s), & M^c(t,s) = 1,\ M^r(t,s) = 0, \\ V^r(t,s), & M^c(t,s) = 0,\ M^r(t,s) = 1, \\ \dfrac{V^c(t,s) + V^r(t,s)}{2}, & M^c(t,s) = 1,\ M^r(t,s) = 1, \\ 0, & M^c(t,s) = 0,\ M^r(t,s) = 0. \end{cases} \tag{9}$$

It is worth emphasizing that $\mathbf{V}^f$ and $\mathbf{M}^f$ are introduced solely to provide an explicit cue of joint observations at the input level. The subsequent source-aware input stem further learns the adaptive integration of CAV and RSU sensing modalities in the feature space through a gated fusion module. By jointly retaining source-specific observations, fused observations, observability information, and explicit positional encoding, the model is able to capture where observations are available, what values are observed, and where each observation is located within the overall spatiotemporal structure.

2.2.2. Source-aware stem

CAV and RSU observations differ substantially in their sampling mechanisms. Specifically, CAV sensing provides sparse mobile observations distributed along vehicle trajectories, whereas RSU sensing offers relatively continuous observations over localized fixed regions. If these two sensing sources are directly stacked and fed into the backbone network, their heterogeneity in coverage pattern, information density, and reliability may be obscured. To address this issue, we introduce a source-aware input stem before the backbone network, where the two sensing modalities are explicitly modeled by separate branches and adaptively fused in a local manner.

More specifically, the CAV speed observations and their corresponding observability mask are first processed by a convolutional branch to produce the CAV feature representation:

$$\mathbf{H}^c = \phi_c\big([\mathbf{V}^c, \mathbf{M}^c]\big). \tag{10}$$

Similarly, the RSU speed observations and mask are processed by another structurally symmetric branch to obtain the RSU feature representation:

$$\mathbf{H}^r = \phi_r\big([\mathbf{V}^r, \mathbf{M}^r]\big). \tag{11}$$

A gated fusion function $g(\cdot)$ is then introduced to generate a local fusion weight based on the two feature representations and their corresponding observability masks:

$$\alpha(t,s) = g\big(\mathbf{H}^c(t,s), \mathbf{H}^r(t,s), M^c(t,s), M^r(t,s)\big). \tag{12}$$

Based on this adaptive weight, the source-aware fused feature representation is defined as

$$\mathbf{H}^f(t,s) = \alpha(t,s)\mathbf{H}^c(t,s) + \big(1 - \alpha(t,s)\big)\mathbf{H}^r(t,s). \tag{13}$$

It is important to note that $\mathbf{H}^f$ denotes the fused representation learned in the feature space through source-specific branch encoding and gated integration. Its role is to capture the adaptive combination of heterogeneous CAV and RSU sensing sources under local spatiotemporal context. In contrast, $\mathbf{V}^f$ and $\mathbf{M}^f$ denote the explicitly fused observation values and the corresponding joint observability mask at the input level, respectively, which provide the network with coarse-grained anchors of joint observations. Therefore, $\mathbf{H}^f$ and $\mathbf{V}^f$ correspond to feature-level fusion and observation-level fusion, respectively, and the information they provide is complementary rather than redundant.

Based on the above design, $\mathbf{H}^f$ is further concatenated with the fused observation $\mathbf{V}^f$, fused mask $\mathbf{M}^f$, temporal coordinate $\mathbf{C}^t$, and spatial coordinate $\mathbf{C}^s$, and then mapped by a hybrid convolution layer into a unified latent representation:

$$\mathbf{H}^0 = \psi\big([\mathbf{H}^f, \mathbf{V}^f, \mathbf{M}^f, \mathbf{C}^t, \mathbf{C}^s]\big). \tag{14}$$

This design enables the model, prior to entering the backbone network, to preserve the adaptively fused representation of heterogeneous sensing sources in the feature space, while simultaneously exploiting explicit joint observations and positional information to provide anchoring and supplementary cues for subsequent full-field traffic state reconstruction.

2.2.3. Encoder–decoder backbone

After the source-aware stem, we adopt a U-Net-style two-dimensional spatiotemporal encoder–decoder backbone for full-field traffic state reconstruction. The encoder progressively enlarges the receptive field through a series of residual convolutional blocks and downsampling operations, so as to capture spatiotemporal traffic-state features at multiple scales. Let the encoder mapping be denoted by $\mathcal{E}(\cdot)$. Then, the resulting latent representation can be written as

$$\mathbf{Z} = \mathcal{E}\left(\mathbf{H}^0\right). \tag{15}$$

Here, $\mathbf{Z}$ denotes the high-level latent representation learned by the encoder. The decoder subsequently restores the time–space resolution through upsampling and skip connections, and outputs the final reconstructed traffic state field. Since speed is adopted in the current implementation as the core representation of traffic state, the decoder output is expressed as

$$\hat{\mathbf{V}}^{(l)} = D(\mathbf{Z}), \qquad \hat{\mathbf{V}}^{(l)} \in \mathbb{R}^{T\times S}. \tag{16}$$

This strategy, which reconstructs the entire time–space state field in a single forward pass, is more suitable than pointwise query-based implicit representation methods for recovering the global morphology of congestion patterns and propagation structures. It also facilitates the formation of multi-scale spatiotemporal organization features within the network.

2.2.4. Block-masked self-supervised objectives

In the traffic state reconstruction task, directly using the complete dense state field as the sole supervision allows the model to learn a mapping from sparse inputs to dense outputs, but it may also cause the model to degenerate into fitting the label distribution, rather than genuinely capturing the propagation structure of traffic states on the time–space plane. To address this issue, we adopt a block-masked self-supervised reconstruction mechanism as the primary training objective. The key idea is to further impose artificially constructed contiguous spatiotemporal missing regions within the originally observable areas of different sensing sources, while retaining the remaining observations as contextual input and requiring the model to recover the originally available observations inside the masked regions.

Specifically, for any sensing source $x \in \{c, r, f\}$, let its observation field and observability mask be denoted by $\mathbf{V}^x$ and $\mathbf{M}^x$, respectively. During training, several temporary spatiotemporal masking blocks are randomly generated within the originally observable region defined by $\mathbf{M}^x$. This yields a target mask $\mathbf{M}^{x,\mathrm{tar}}$ and a context mask $\mathbf{M}^{x,\mathrm{ctx}}$, which satisfy

$$\mathbf{M}^{x,\mathrm{ctx}} + \mathbf{M}^{x,\mathrm{tar}} \le \mathbf{M}^x, \qquad \mathbf{M}^{x,\mathrm{ctx}} \odot \mathbf{M}^{x,\mathrm{tar}} = 0. \tag{17}$$

Accordingly, the retained contextual input and the masked target region are defined as

$$\mathbf{V}^{x,\mathrm{ctx}} = \mathbf{V}^x \odot \mathbf{M}^{x,\mathrm{ctx}}, \qquad \mathbf{V}^{x,\mathrm{tar}} = \mathbf{V}^x \odot \mathbf{M}^{x,\mathrm{tar}}. \tag{18}$$

Compared with random pointwise masking, block masking better matches realistic traffic sensing failures, such as local communication interruption, temporary continuous missing segments, or RSU coverage blind zones. Therefore, it encourages the model to infer contiguous missing regions from both neighboring local context and broader global context, which is more beneficial for learning the spatiotemporal continuity and propagation structure of traffic states.

On this basis, we mainly construct masked reconstruction losses for the two source-specific observations, namely CAV and RSU observations, which are defined as

$$\mathcal{L}_{\mathrm{cav}} = \frac{\left\| \left(\hat{\mathbf{V}} - \mathbf{V}^c\right) \odot \mathbf{M}^{c,\mathrm{tar}} \right\|_2^2}{\sum \mathbf{M}^{c,\mathrm{tar}} + \epsilon}, \tag{19}$$

$$\mathcal{L}_{\mathrm{rsu}} = \frac{\left\| \left(\hat{\mathbf{V}} - \mathbf{V}^r\right) \odot \mathbf{M}^{r,\mathrm{tar}} \right\|_2^2}{\sum \mathbf{M}^{r,\mathrm{tar}} + \epsilon}. \tag{20}$$

In addition, for locations where both CAV and RSU observations are simultaneously available, we further introduce a cross-source consistency loss to reduce local discrepancies between different sensing sources. The overlap mask is defined as

$$\mathbf{M}^o = \mathbf{M}^c \odot \mathbf{M}^r, \tag{21}$$

and the corresponding consistency loss is given by

$$\mathcal{L}_{\mathrm{cons}} = \frac{\left\| \left(\hat{\mathbf{V}} - \frac{\mathbf{V}^c + \mathbf{V}^r}{2}\right) \odot \mathbf{M}^o \right\|_2^2}{\sum \mathbf{M}^o + \epsilon}. \tag{22}$$

Accordingly, the self-supervised training adopted in this study is not merely a missing-value imputation scheme. Rather, through structured masking recovery and cross-source consistency constraints, it enables the model to jointly learn local contextual inference, multi-source information integration, and the spatiotemporal continuity of traffic states.

2.3. Physics-guided regularization

Relying solely on self-supervised reconstruction losses often leads the model to generate traffic fields that are numerically smooth but structurally weak. Since traffic states are not ordinary images but spatiotemporal physical fields governed by traffic flow dynamics, we further introduce a set of physics-guided regularization terms to improve the plausibility of the reconstructed field in terms of wave propagation, congestion organization, and directional smoothness. Because speed is adopted as the state variable in the current implementation, all regularization terms are applied to the reconstructed speed field $\hat{x}(t,s) = \hat{v}(t,s)$.

In implementation, all temporal and spatial derivatives, as well as directional gradients, are approximated using finite differences on the discretized time–space grid. Unless otherwise specified, Ω denotes the set of valid grid cells in the reconstructed time–space domain, and $|\Omega|$ denotes the number of grid cells in Ω.

First, a *state-aware wave propagation constraint* is introduced to encourage the predicted speed field to evolve consistently with local traffic-wave dynamics. Since traffic perturbations propagate at different speeds under free-flow and congested conditions, a congestion probability field is constructed from the predicted speed as

$$P_c(t,s) = \sigma\left(\frac{v_{th} - \hat{v}(t,s)}{\delta}\right), \tag{23}$$

where $\sigma(\cdot)$ denotes the sigmoid function, v_{th} is a congestion threshold, and δ is a smoothing coefficient controlling the sharpness of the transition between free-flow and congested states. The local propagation speed is then defined by interpolating between the backward congestion wave speed w_b and the free-flow propagation speed w_f:

$$w(t,s) = P_c(t,s)\, w_b + \big(1 - P_c(t,s)\big)\, w_f. \tag{24}$$

Accordingly, the wave propagation residual is defined as

$$\mathcal{R}_{wave}(t,s) = \frac{\partial \hat{v}(t,s)}{\partial t} + w(t,s)\frac{\partial \hat{v}(t,s)}{\partial s}, \tag{25}$$

This term is introduced as a propagation-aware regularization term to enforce physically plausible directional evolution of the reconstructed traffic state in the time-space plane. The corresponding loss is defined as

$$\mathcal{L}_{wave} = \frac{1}{|\Omega|}\sum_{(t,s)\in\Omega} \mathcal{R}_{wave}(t,s)^2. \tag{26}$$

Second, to preserve the continuous band-like structure of low-speed congestion regions in the time–space plane, we impose a congestion continuity constraint on the congestion probability field. Unlike $\mathcal{L}_{wave}$, which acts directly on the predicted speed field, this term explicitly regularizes the structural continuity of congestion patterns. The congestion-wave residual is defined as

$$\mathcal{R}_{cong}(t,s) = \frac{\partial P_c(t,s)}{\partial t} + w_b\frac{\partial P_c(t,s)}{\partial s}, \tag{27}$$

with the corresponding loss

$$\mathcal{L}_{cong} = \frac{1}{|\Omega|}\sum_{(t,s)\in\Omega} \mathcal{R}_{cong}(t,s)^2. \tag{28}$$

Third, to prevent isotropic smoothing from excessively blurring wavefront boundaries, we introduce an anisotropic smoothing constraint. Let $\nabla_{\parallel}\hat{v}$ and $\nabla_{\perp}\hat{v}$ denote the directional derivatives of $\hat{v}$ along and orthogonal to the local propagation direction, respectively. The local propagation direction is determined by the wave speed field $w(t,s)$, and both directional gradients are implemented as directional finite-difference operators on the grid. The anisotropic smoothness loss is defined as

$$\mathcal{L}_{aniso} = \lambda_{\parallel}\|\nabla_{\parallel}\hat{v}\|_1 + \lambda_{\perp}\|\nabla_{\perp}\hat{v}\|_1, \qquad \lambda_{\parallel} > \lambda_{\perp}, \tag{29}$$

where $\lambda_{\parallel}$ and $\lambda_{\perp}$ are directional smoothness weights along and across the propagation direction, respectively. Here, $\|\cdot\|_1$ denotes the element-wise L_1 norm computed over the corresponding grid domain. This term encourages stronger consistency along the propagation direction while preserving sharper gradients and boundaries across the orthogonal direction.

In addition, a weak conservation-based continuity regularization and a total variation regularization are incorporated. Let $\rho(t,s)$ and $q(t,s)$ denote the density and flow approximately inferred from the predicted speed field through a predefined fundamental-diagram-based mapping. In the current implementation, a simplified Greenshields-type mapping is adopted as an auxiliary approximation:

$$\rho(t,s) = \rho_j\left(1 - \frac{\hat{v}(t,s)}{v_f}\right), \qquad q(t,s) = \rho(t,s)\hat{v}(t,s), \tag{30}$$

where ρ_j and v_f denote the jam density and free-flow speed, respectively. This mapping is introduced only to provide a weak conservation-oriented regularization and is not intended to serve as the primary traffic flow model. The continuity residual is defined as

$$\mathcal{R}_{cont}(t,s) = \frac{\partial \rho(t,s)}{\partial t} + \frac{\partial q(t,s)}{\partial s}, \tag{31}$$

with the corresponding continuity loss

$$\mathcal{L}_{cont} = \frac{1}{|\Omega|}\sum_{(t,s)\in\Omega} \mathcal{R}_{cont}(t,s)^2. \tag{32}$$

To further suppress local oscillations while preserving the overall organization of the reconstructed field, a total variation regularization is introduced:

$$\mathcal{L}_{tv} = \frac{1}{2}\left(\|\hat{\mathbf{V}}_{t+1,s} - \hat{\mathbf{V}}_{t,s}\|_1 + \|\hat{\mathbf{V}}_{t,s+1} - \hat{\mathbf{V}}_{t,s}\|_1\right), \tag{33}$$

where $\hat{\mathbf{V}}_{t+1,s}$ and $\hat{\mathbf{V}}_{t,s+1}$ denote adjacent grid values of the reconstructed speed field along the temporal and spatial dimensions, respectively, and the L_1 norm is computed element-wise over all valid neighboring grid pairs.

Finally, the overall training objective is formulated as

$$\mathcal{L}_{total} = \lambda_c\mathcal{L}_{cav} + \lambda_r\mathcal{L}_{rsu} + \lambda_{cs}\mathcal{L}_{cons} + \lambda_w\mathcal{L}_{wave} + \lambda_{cw}\mathcal{L}_{cong} + \lambda_a\mathcal{L}_{aniso} + \lambda_{ct}\mathcal{L}_{cont} + \lambda_{tv}\mathcal{L}_{tv}, \tag{34}$$

where λ_c, λ_r, λ_{cs}, λ_w, λ_{cw}, λ_a, λ_{ct}, and λ_{tv} are nonnegative weighting coefficients used to balance the source-specific reconstruction terms, cross-source consistency term, and physics-guided regularization terms.

The above objective jointly accounts for source-specific reconstruction accuracy, cross-source consistency, and the structural plausibility of traffic state propagation.

3. Experimental design

This section presents an experimental validation of the proposed traffic state reconstruction method under multi-source partial observations. It first introduces the experimental data, observation construction procedure, training and testing settings, benchmark methods, evaluation metrics, and implementation details. Comparative experiments are then conducted under different CAV penetration rates, across different lanes, and at the corridor-wide level, with particular attention to the model's ability to capture the propagation structure of traffic congestion. Finally, ablation studies are carried out to examine the specific effects of block-wise masking, the self-supervised learning mechanism, and traffic-flow physics-informed priors on speed-field reconstruction performance.

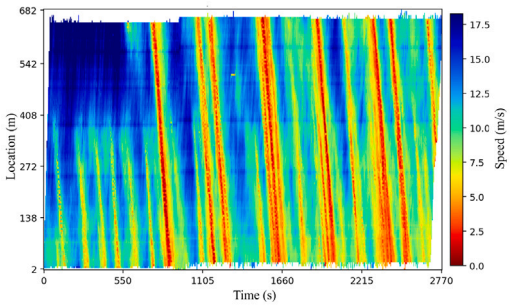
3.1. Data construction and experimental design

The experiments in this study are conducted using original FHWA NGSIM US-101 trajectory dataset. The US-101 data were collected on the southbound US-101 freeway in Los Angeles, California, on June 15, 2005, using synchronized video cameras, and vehicle trajectories were extracted by the NG-VIDEO software at a temporal resolution of 0.1 s. In this study, the original NGSIM trajectory records were used as the primary data source and were further processed only for field standardization, unit conversion, duplicate removal, trajectory cleaning and segmentation, and the construction of CAV/RSU sparse observations and traffic-state grids. No third-party reconstructed or smoothed NGSIM trajectory dataset was used [32]. This dataset provides high-resolution microscopic vehicle trajectories over continuous time within the study corridor and is able to comprehensively reflect vehicle positions, speeds, accelerations, and lane-changing behaviors during operation. Since this study focuses on reconstructing corridor-level continuous traffic speed fields under partially observable conditions, the raw trajectory data are first subjected to unified preprocessing, upon which a spatiotemporal representation suitable for multi-source observation fusion is constructed.

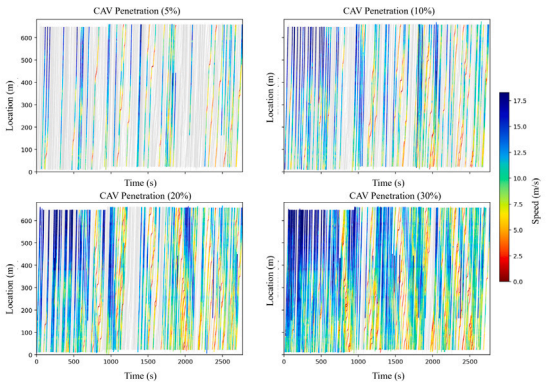
After completing field mapping, unit normalization, missing-value processing, trajectory sorting, and abnormal-record cleaning for the raw trajectory data, a standardized trajectory file is obtained. Based on these standardized trajectories, multiple categories of data are further generated for traffic-state reconstruction and performance evaluation. First, complete traffic state data are aggregated from the full trajectory information and used as the reference ground truth for offline evaluation. Second, CAV trajectory observations are constructed through random sampling of vehicle trajectories so as to simulate sparse mobile observations under different CAV penetration rates. In addition, RSU observations are generated by defining fixed spatial coverage zones within a unified longitudinal coordinate system, thereby representing traffic state observations under localized roadside sensing conditions. Through these procedures, mobile and fixed observations are organized into a consistent spatiotemporal framework, laying the foundation for subsequent multi-source fusion and reconstruction.

To balance the ability to represent traffic-state evolution details with the computational tractability of model training, a regular spatiotemporal grid with a resolution of $5\ \mathrm{s} \times 5\ \mathrm{m}$ is adopted as the primary reconstruction scale. Specifically, the time axis is discretized at 5-second intervals, while the spatial axis is partitioned every 5 m along the unified longitudinal coordinate s, yielding a two-dimensional time-space speed field to represent the traffic state. Compared with local regression schemes based on single-point queries, this regular-grid representation is more conducive to characterizing the continuous propagation structure of traffic congestion waves in the time-space plane.

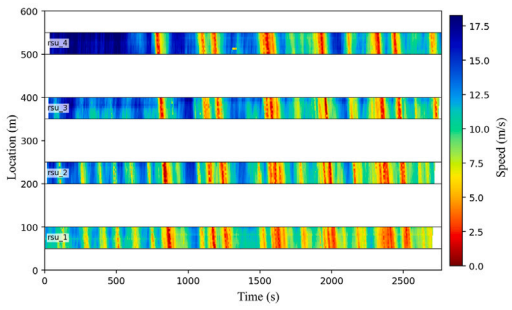
To evaluate the applicability of the model under different observation coverage conditions, four CAV penetration scenarios are considered, namely 5%, 10%, 20%, and 30%. First, a continuous traffic speed field on the regular spatiotemporal grid is constructed from the complete trajectory data and used as the reference ground truth for subsequent sparse-observation generation and reconstruction evaluation. Then, under each penetration scenario, a corresponding proportion of vehicles is randomly selected from the standardized trajectory data to generate CAV trajectory observations with different levels of sparsity. Meanwhile, RSU observations are constructed through predefined spatial coverage regions, providing relatively stable fixed observations for local areas. In this study, the RSU placement was not optimized but predefined to provide uniformly distributed local observations along the corridor. The purpose of this configuration was to evaluate the proposed reconstruction framework under sparse but regularly distributed roadside sensing conditions, rather than to investigate the optimal deployment of RSUs. Within each covered segment,



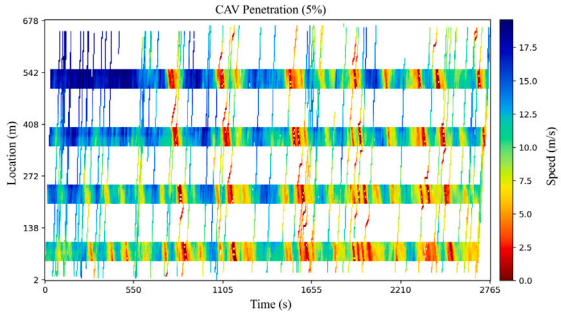
(a) Complete time-space speed field under the 5 s × 5 m grid resolution.



(b) Sparse CAV trajectory observations under different probe penetration rates (5%, 10%, 20%, and 30%).



(c) RSU sparse coverage field under the 5 s × 5 m grid resolution.



(d) Fused sparse observation field obtained by combining CAV and RSU observations under the 5% penetration scenario.

Fig. 3. Illustration of the complete traffic state and the construction of multi-source sparse observations.

traffic speed observations were assumed to be continuously available over time and were mapped onto the regular 5 s × 5 m time-space grid. Fig. 3 illustrates the complete traffic state field, the distribution of mobile observations under different CAV penetration rates, the fixed coverage pattern of RSU observations, and an example of the fused sparse observation field obtained by combining CAV and RSU observations under the 5% penetration scenario. It can be observed that the complete traffic state field clearly reveals the evolution of speed disturbances and low-speed band structures in the time-space plane. In contrast, CAV observations form sparse line-like samples along vehicle trajectories, and their spatiotemporal coverage gradually improves as the penetration rate increases. Compared with mobile observations, RSU observations mainly provide continuous temporal coverage over several fixed spatial intervals, making them more suitable for offering stable local-state references. When the two observation sources are combined, the resulting fused sparse field exhibits both trajectory-aligned mobile samples and locally continuous roadside coverage, providing a more informative yet still incomplete observation basis for reconstruction. Overall, the two observation sources reflect the distinct characteristics of mobile sensing and roadside sensing, respectively, and complement each other under a unified spatiotemporal grid representation, thereby jointly supporting the recovery of continuous traffic states.

3.2. Benchmark methods and evaluation metrics

To comprehensively evaluate the performance of the proposed method in the task of traffic-state reconstruction under multi-source partial observations, several representative benchmark methods are constructed from the perspectives of local modeling, global implicit representation, supervised reconstruction, and self-supervised physics-enhanced reconstruction.

The benchmark methods considered in this study include several representative reconstruction paradigms.

The first benchmark is a *local query-based reconstruction model*. This model centers on a target spatiotemporal query point, encodes local CAV observations and RSU neighborhood statistics to form a query-relevant representation, and outputs the target-point speed through an implicit decoder. Although this type of model exhibits strong local fitting capability, it often struggles to preserve the continuity of congestion-wave propagation structures over the global spatiotemporal domain.

The second benchmark is a *single-source spatiotemporal implicit neural representation (ST-INR) baseline*. This method formulates traffic-state reconstruction as a matrix recovery problem on a regular spatiotemporal grid and learns the entire speed field through

a global implicit representation using a single sparse observation source. Compared with local query-based methods, it is more favorable for capturing the overall propagation pattern of traffic states in the time-space plane.

The third benchmark is a *multi-source spatiotemporal implicit neural representation (ST-INR) baseline*. Compared with the single-source ST-INR model, this variant incorporates both CAV observations and RSU observations within the same global implicit reconstruction framework, and is used to examine whether a direct multi-source extension can further improve reconstruction performance.

The fourth benchmark is a *supervised multi-channel grid fusion model*. This model takes sparse CAV observations, local RSU observations, observation masks, and normalized spatiotemporal coordinates as unified inputs, and directly learns the mapping from multi-source sparse observations to continuous traffic states under dense supervision. Its results can be regarded as a reference upper bound.

The fifth benchmark is a *basic self-supervised grid fusion model*. This model does not rely on dense ground truth and is mainly trained with masked-observation recovery loss and cross-source consistency constraints.

Finally, the proposed method is a *physics-enhanced self-supervised model*. Building upon the basic self-supervised framework, it further incorporates traffic-flow-inspired prior knowledge, including block-wise spatiotemporal masking, state-aware propagation constraints, and anisotropic smoothing constraints, to reconstruct continuous speed fields from multi-source sparse observations without directly relying on dense ground truth.

On this basis, this study further focuses on a *self-supervised reconstruction framework* that does not rely on dense ground truth. The basic self-supervised model is mainly trained with masked-observation recovery loss and cross-source consistency constraints. The physics-enhanced self-supervised model further incorporates traffic-flow prior knowledge, introducing state-aware propagation constraints and anisotropic smoothing constraints to strengthen the learning of congestion propagation structures. The final model adopted in this study is the physics-enhanced self-supervised version, which combines block-wise spatiotemporal masking, state-aware propagation constraints, and anisotropic smoothing constraints to reconstruct continuous speed fields from multi-source sparse observations without directly relying on dense ground truth.

To quantitatively evaluate the reconstruction performance of different methods, mean absolute error (MAE), root mean square error (RMSE), and mean absolute percentage error (MAPE) are adopted as the primary evaluation metrics. Let v_i denote the reference ground-truth speed and $\hat{v}_i$ denote the predicted speed at the ith spatiotemporal grid, with N being the total number of evaluated grids. These metrics are defined as

$$\mathrm{MAE} = \frac{1}{N}\sum_{i=1}^{N}\left|\hat{v}_i - v_i\right|, \tag{35}$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left(\hat{v}_i - v_i\right)^2}, \tag{36}$$

$$\mathrm{MAPE} = \frac{100\%}{N}\sum_{i=1}^{N}\left|\frac{\hat{v}_i - v_i}{v_i + \epsilon}\right|, \tag{37}$$

where ϵ is a small constant introduced to avoid numerical instability when the ground-truth speed is close to zero. MAE and RMSE characterize the absolute reconstruction error, with RMSE being more sensitive to large deviations, while MAPE reflects the relative prediction error. In addition to quantitative metrics, visualized traffic-state fields are also examined to assess whether the model can correctly recover the formation, diffusion, and propagation of congestion, with particular attention paid to the continuity and propagation direction of low-speed congestion bands in the t–s plane.

3.3. Computational complexity and inference efficiency

To address the computational feasibility of the proposed framework, we further evaluated the model complexity and inference efficiency on the US-101 test case. The final grid-fusion network contains 2.13 million trainable parameters. Since the model is implemented as a fully convolutional spatiotemporal grid reconstruction network, its inference complexity grows approximately linearly with the number of time-space grid cells for a fixed channel width and network depth.

The inference benchmark was conducted on CPU using the lane-level grids under the 10% CAV penetration setting. For each lane, 10 warm-up runs and 50 repeated forward passes were performed. As shown in Table 1, the average inference time per lane was 0.086 s, and the estimated inference time for reconstructing all eight lanes of the corridor was 0.687 s. The model processed approximately 5.28×10^5 grid cells per second on average. These results indicate that the proposed framework is computationally lightweight and suitable for large-scale corridor-level deployment, especially since lane-level inference can be parallelized in practical implementations.

4. Experimental results and analysis

4.1. Comparison of different methods

To verify the effectiveness of the proposed method, a comparative study was first conducted under the 10% CAV penetration scenario. The results are reported in Table 2. As shown in the table, the reconstruction accuracy of the different models exhibits

Table 1
Inference efficiency of the proposed model under 10% CAV penetration.

Lane ID	Grid cells	Mean time (s)	P95 time (s)
1	74 655	0.115	0.121
2	75 344	0.117	0.122
3	75 480	0.121	0.129
4	75 761	0.127	0.135
5	75 208	0.120	0.128
6	23 460	0.045	0.047
7	6894	0.023	0.024
8	5073	0.020	0.021
Average per lane	–	0.086	–
Full corridor estimate	411 875	0.687	–

Table 2
Reconstruction accuracy of different methods.

Method	MAE (m/s)	RMSE (m/s)
Local query-based reconstruction model	2.511	3.335
Single-source ST-INR baseline	1.729	2.409
Multi-source ST-INR baseline	1.793	2.469
Supervised multi-channel grid fusion model	0.507	0.671
Basic self-supervised model	1.298	1.839
Proposed model	0.781	1.046

substantial variation. The local query-based reconstruction model achieved an MAE of 2.511 m/s and an RMSE of 3.335 m/s, indicating that although this class of methods can exploit local neighborhood information to regress the target point, it has clear limitations in recovering the overall propagation structure of the traffic state field. In contrast, the single-source ST-INR baseline, which adopts global modeling on a regular grid, reduced the MAE and RMSE to 1.729 m/s and 2.409 m/s, respectively. This suggests that reformulating the reconstruction task as a global completion problem on a regular spatiotemporal grid is more conducive to capturing the overall structural characteristics of traffic-wave propagation.

Under the multi-source input setting, the multi-source ST-INR variant obtained an MAE of 1.793 m/s and an RMSE of 2.469 m/s. Its performance did not further improve relative to the single-source ST-INR baseline, indicating that under the current fusion strategy, the multi-source observations were not yet fully exploited and may even have introduced a certain degree of inter-source conflict. The supervised multi-channel grid fusion model further achieved an MAE of 0.507 m/s and an RMSE of 0.671 m/s, substantially outperforming all of the aforementioned baselines. This result demonstrates that when the complete speed field is available as the supervisory signal, the global fusion framework based on multi-channel grid representation has a markedly stronger capacity for speed-field reconstruction. It should be noted, however, that the supervised model relies on dense ground truth during training; therefore, it is more appropriately regarded as an upper-bound reference rather than a fully fair comparison target for the self-supervised setting with purely sparse observations.

Without relying on dense supervision, multiple self-supervised models were further compared. Under the 10% CAV penetration scenario, the basic self-supervised model achieved an MAE of 1.298 m/s and an RMSE of 1.839 m/s. By contrast, the proposed model, which incorporates block-wise spatiotemporal masking and state-aware propagation constraints, further improved the reconstruction accuracy to an MAE of 0.781 m/s and an RMSE of 1.046 m/s. These results indicate that the proposed physics-enhanced self-supervised framework can achieve better reconstruction performance than the basic self-supervised baseline without relying on dense ground truth.

4.2. Overall reconstruction performance under different CAV penetration rates

Building upon the preceding comparison of different methods, this study further fixes the proposed physics-enhanced self-supervised model and evaluates its overall reconstruction capability under different CAV penetration rates, with the aim of analyzing the impact of observation coverage on model performance. Table 3 presents the reconstruction results of the proposed model under four CAV penetration scenarios, namely 5%, 10%, 20%, and 30%.

As shown in Table 3, as the CAV penetration rate gradually increases from 5% to 30%, the overall reconstruction error of the proposed model exhibits a relatively stable downward trend. Specifically, when the CAV penetration rates are 5%, 10%, 20%, and 30%, the corridor-level MAE values are 1.000 m/s, 0.781 m/s, 0.691 m/s, and 0.666 m/s, respectively; the corresponding RMSE values are 1.374 m/s, 1.046 m/s, 0.920 m/s, and 0.896 m/s; and the MAPE values are 12.34%, 9.61%, 8.57%, and 8.38%, respectively. These results indicate that, under the same model architecture and training framework, increasing the CAV penetration rate can provide more sufficient mobile observations for speed-field reconstruction, thereby effectively improving the overall reconstruction accuracy.

From the perspective of relative improvement, compared with the 5% penetration scenario, the RMSE under the 30% penetration scenario decreases from 1.374 m/s to 0.896 m/s, corresponding to a reduction of approximately 34.8%. This demonstrates that

Table 3
Reconstruction results of the proposed model under different CAV penetration rates.

CAV penetration rate	MAE (m/s)	RMSE (m/s)	MAPE (%)
5%	1.000	1.374	12.34
10%	0.781	1.046	9.61
20%	0.691	0.920	8.57
30%	0.666	0.896	8.38

improving mobile observation coverage can substantially enhance traffic-state reconstruction capability. However, when the CAV penetration rate further increases from 20% to 30%, the RMSE decreases only slightly from 0.920 m/s to 0.896 m/s, indicating a much weaker improvement. This suggests that the performance gain of the model begins to exhibit a diminishing marginal return in the high-penetration regime. In other words, once the mobile observation coverage reaches a certain level, further increasing the number of CAVs can still improve traffic-state reconstruction quality, but the incremental benefit gradually decreases.

4.3. Lane-level heterogeneity and propagation visualization analysis

Beyond the overall reconstruction results, the lane-level results exhibit clear heterogeneity. The lane-level reconstruction performance is summarized in Table 4. As can be observed from the table, lanes 3 and 4 generally show relatively stable reconstruction performance, achieving comparatively low errors under multiple penetration-rate scenarios. For example, under the 10% CAV penetration scenario, lane 3 attains an RMSE of 1.227 m/s; under the 20% scenario, lane 4 achieves an RMSE of 1.152 m/s; and under the 30% scenario, lane 3 yields an RMSE of 1.195 m/s. These results suggest that the traffic-state evolution patterns in some middle mainline lanes are relatively stable, and that the multi-source observations provide sufficiently informative constraints on their spatiotemporal structures, enabling the model to achieve favorable reconstruction performance.

In contrast, lane 7 consistently remains the most challenging lane to reconstruct. Its RMSE values under 5%, 10%, 20%, and 30% CAV penetration rates reach 15.217 m/s, 4.304 m/s, 6.378 m/s, and 3.690 m/s, respectively, which are substantially higher than those of the other lanes. Meanwhile, lanes 6 and 8 also exhibit relatively large errors. For instance, under the 30% penetration scenario, the RMSE values of lanes 6 and 8 are 2.940 m/s and 3.074 m/s, respectively. These findings indicate that different lanes differ markedly in observation coverage, traffic-state complexity, and propagation patterns, such that a unified model does not face the same reconstruction difficulty across all lanes. In particular, for edge lanes or lanes with more pronounced state variations, insufficient local observations and more complex propagation patterns jointly amplify the reconstruction difficulty. More specifically, the high error on lane 7 is likely related to three factors. First, the effective CAV observation density may be lower or more uneven on this lane because CAV observations are generated through random trajectory sampling. Although RSU observations provide fixed local coverage, they cannot fully compensate for missing observations outside the covered regions. Second, lane 7 may experience stronger traffic-state instability, including speed fluctuations, localized disturbances, and boundary-lane or lane-changing effects, leading to sharper and less regular speed-field variations. Third, the current framework reconstructs each lane independently and does not explicitly capture lateral interactions among adjacent lanes. Therefore, the large error on lane 7 should be interpreted as the combined result of observation sparsity, unstable traffic dynamics, and the limitation of the lane-independent modeling design.

From the perspective of average lane-level results, the mean RMSE values under the 5%, 10%, 20%, and 30% penetration scenarios are 3.675 m/s, 2.163 m/s, 2.418 m/s, and 2.017 m/s, respectively. Overall, penetration rates of 10% and above can substantially reduce the average reconstruction error. However, due to the influence of a few particularly difficult lanes, the average RMSE under the 20% scenario is slightly higher than that under the 10% scenario, indicating that local extreme errors exert a strong influence on lane-level averages. This also suggests that assessing model performance solely from corridor-level aggregate results is insufficient, and that lane-level analysis is necessary for a more detailed evaluation of model applicability and robustness.

To further examine the capability of different methods in characterizing congestion propagation structures, the *Basic self-supervised model*, the *Multi-source ST-INR baseline*, and the *Proposed model* are selected for visual comparison, as shown in Fig. 4. The *Local query-based reconstruction model*, the *Single-source ST-INR baseline*, and the *Supervised multi-channel grid fusion model* are not included in this visualization for the following reasons. First, the local query-based model mainly focuses on pointwise regression from local neighborhoods and is less suitable for revealing representative large-scale spatiotemporal propagation structures. Second, the single-source ST-INR baseline does not explicitly exploit multi-source observations, whereas the core problem considered in this study is traffic-state reconstruction under multi-source partial observations; therefore, its multi-source counterpart is more relevant for comparison. Third, the supervised multi-channel grid fusion model relies on dense ground truth during training and is thus better regarded as an upper-bound reference than as a directly comparable method under the same self-supervised setting.

As can be seen from Fig. 4, the proposed model is able to recover the major distribution patterns of both free-flow and congested regions in the time-space plane, while also providing a clearer depiction of congestion-wave propagation trajectories over time and space. By contrast, although the basic self-supervised model can recover some low-speed regions, it still shows noticeable deficiencies in the continuity and structural integrity of congestion bands, with visible fragmentation, local blurring, and less stable propagation directions. The multi-source ST-INR baseline performs better than the basic self-supervised model in recovering the overall contour of the traffic state, but its reconstruction still suffers from local noise and structural discontinuities, particularly in the delineation of congestion-band boundaries and fine-grained propagation textures.

Table 4
Lane-level reconstruction results under different CAV penetration rates.

CAV penetration rate (%)	Lane ID	MAE (m/s)	RMSE (m/s)	MAPE (%)
5	1	1.301336765	1.695970654	70.63156366
5	2	1.222306252	1.582332015	43.17235947
5	3	1.223624825	1.588968635	32.98044801
5	4	1.059328079	1.381068110	52.91078091
5	5	1.129855514	1.526451111	22.70183116
5	6	2.114799738	2.770677328	15.15719891
5	7	14.886164670	15.217092510	119.96247770
5	8	2.722524405	3.639967918	18.63301694
10	1	1.145358324	1.478305101	61.53541207
10	2	1.197159767	1.531404495	44.59907115
10	3	0.935263991	1.227428198	20.53687721
10	4	0.986779571	1.303646564	50.94919801
10	5	1.051768422	1.418015480	20.94137967
10	6	2.095717430	2.785798788	14.96643871
10	7	3.350331545	4.304294109	29.17877138
10	8	2.468692780	3.258638859	16.57256484
20	1	1.150389910	1.474464297	71.67367339
20	2	1.009701967	1.309052825	44.20729876
20	3	0.989572048	1.297069907	35.19448042
20	4	0.884037197	1.151939154	56.94493055
20	5	1.004334569	1.399503469	23.35768342
20	6	2.323296309	3.058663607	15.83414823
20	7	5.656099796	6.378263950	38.66631091
20	8	2.490822554	3.275361300	16.18830562
30	1	1.050914288	1.390743375	61.21351719
30	2	1.050596356	1.356806755	43.34339797
30	3	0.899128854	1.195457101	44.92662549
30	4	0.930072069	1.216926813	62.46446371
30	5	0.955381632	1.268612385	25.19947588
30	6	2.259058475	2.939921856	15.91954529
30	7	3.016537666	3.689841509	22.63864875
30	8	2.348487854	3.073725224	15.82047790

Table 5
Ablation results under lane 1 with a 10% CAV penetration rate.

Model Variant	MAE (m/s)	RMSE (m/s)	MAPE (%)
Supervised multi-channel grid fusion	0.507	0.671	27.05
Basic self-supervised (point masking)	1.298	1.839	71.47
Block-masked self-supervised (no physics)	1.151	1.332	62.94
Proposed physics-enhanced self-supervised model	0.989	1.332	39.17

In comparison, the proposed model exhibits superior continuity of low-speed congestion bands, substantially alleviating band fragmentation and better preserving the spatial extension and directional consistency of congestion structures in the time-space plane. As illustrated in Fig. 4, the reconstructed low-speed bands produced by the proposed model are more consistent with the ground truth in terms of overall inclination, band continuity, and local propagation patterns.

4.4. Ablation study

To further investigate the roles of the key components in the proposed framework, ablation experiments were conducted from three aspects, namely the self-supervised learning mechanism, the observation masking strategy, and the traffic-flow-informed physical constraints. All ablation experiments were performed under the setting of lane 1 with a 10% CAV penetration rate to ensure direct comparability among different model variants.

As shown in Table 5, the supervised multi-channel grid fusion model achieves the lowest reconstruction errors, with an MAE of 0.507 m/s, an RMSE of 0.671 m/s, and a MAPE of 27.05%. This is expected because the supervised model is trained using dense ground-truth speed fields and therefore can be regarded as an upper-bound reference. In contrast, the basic self-supervised model with point-wise masking obtains an MAE of 1.298 m/s, an RMSE of 1.839 m/s, and a MAPE of 71.47%. Although its accuracy is lower than that of the fully supervised model, it demonstrates the feasibility of reconstructing traffic states from sparse multi-source observations without relying on dense supervision.

The comparison between the basic self-supervised model and the block-masked self-supervised variant further demonstrates the effectiveness of structured masking. After replacing point-wise random masking with block-wise spatiotemporal masking, the RMSE decreases from 1.839 m/s to 1.332 m/s, indicating a substantial improvement in reconstruction stability. This result suggests that

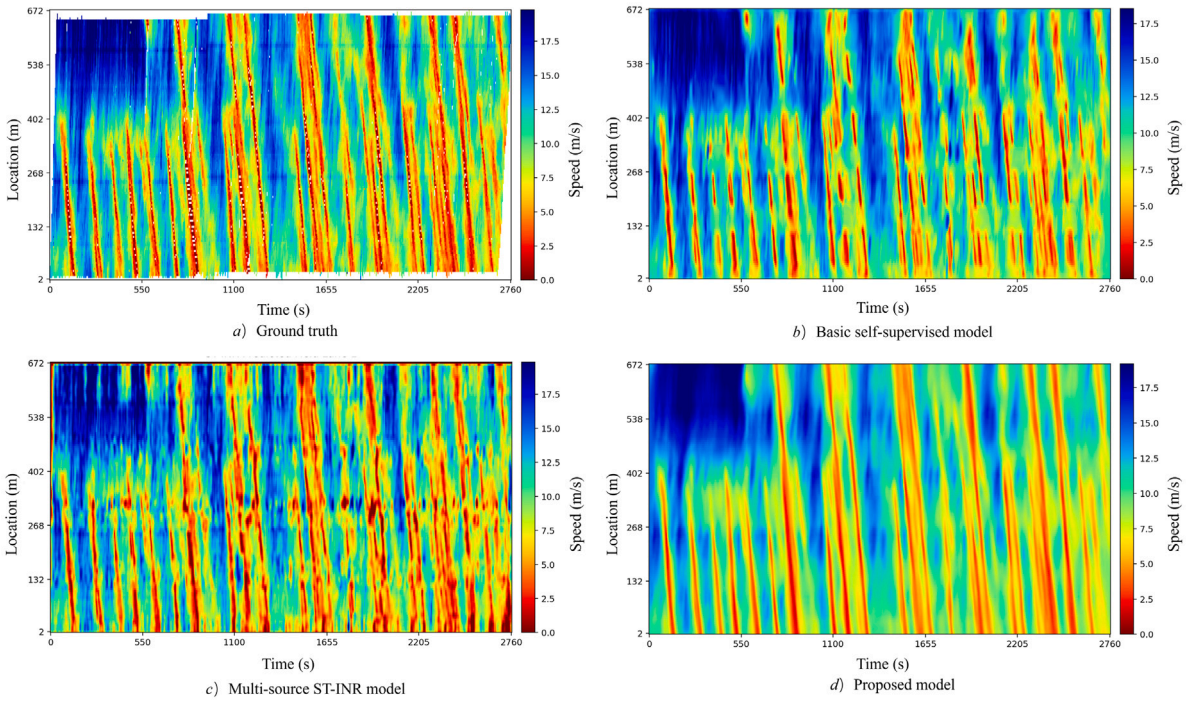


Fig. 4. Visual comparison of traffic-state reconstruction results under the lane 1, 10% CAV penetration scenario.

block-wise masking better matches realistic continuous missing-observation patterns in traffic sensing. By requiring the model to recover contiguous missing regions from surrounding contextual observations, the block-masking strategy encourages the network to learn spatiotemporal propagation structures rather than simply fitting isolated missing points.

The contribution of the traffic-flow-informed physical constraints can be observed by comparing the block-masked self-supervised model without physics and the proposed physics-enhanced self-supervised model. After incorporating the physical regularization terms, the MAE decreases from 1.151 m/s to 0.989 m/s, and the MAPE decreases from 62.94% to 39.17%, while the RMSE remains at 1.332 m/s. These results indicate that the physical constraints improve the overall reconstruction quality, especially by reducing the average absolute error and relative error. The unchanged RMSE also suggests that the physical regularization mainly improves the general consistency of the reconstructed speed field rather than substantially altering a few large-error grid cells.

This improvement can be explained from the perspective of traffic-state propagation. The state-aware propagation constraint encourages the reconstructed speed field to evolve consistently with the local traffic-wave direction, while the anisotropic smoothing constraint helps preserve continuity along the propagation direction without excessively blurring sharp speed transitions across wavefronts. As a result, the model is better able to recover continuous low-speed congestion bands and directionally consistent propagation structures in the time-space plane. Therefore, the physical constraints enhance not only the structural plausibility of the reconstructed traffic field, but also the numerical reconstruction accuracy in terms of MAE and MAPE.

Overall, the ablation results confirm the effectiveness of the major components of the proposed framework. The self-supervised learning mechanism provides the basic capability for reconstructing traffic states from sparse observations without dense ground truth. The block-wise masking strategy significantly improves reconstruction stability under continuous missing patterns. The traffic-flow-informed physical constraints further reduce MAE and MAPE while maintaining comparable RMSE, indicating improved overall reconstruction quality and better physical consistency of the reconstructed traffic field.

To further examine the robustness of the proposed physics-guided design and to clarify the influence of different regularization terms, a sensitivity analysis was conducted for the major physics-guided regularization weights.

To examine the robustness of the proposed framework with respect to major physics-guided hyperparameters, a sensitivity analysis was conducted under lane 1 with a 10% CAV penetration rate. For computational efficiency, all sensitivity experiments were trained for 300 epochs using the same setting, and one regularization weight was varied at a time while the remaining weights were kept fixed. It should be noted that the results in Table 6 are intended to compare the relative sensitivity of different physics-guided weights under the same training budget, rather than to directly replace the main experimental results reported in Table 5, where the final model was trained under the full experimental setting.

As shown in Table 6, the model shows relatively stable performance under moderate perturbations of λ_{cw}, λ_a, and λ_{tv}. For example, varying λ_{cw} from 0.09 to 0.36 leads to RMSE values between 1.522 and 1.538 m/s, while varying λ_a from 0.025 to 0.10 results in RMSE values between 1.473 and 1.498 m/s. Similarly, changing λ_{tv} from 0.004 to 0.016 only changes the RMSE from 1.423 to 1.460 m/s, indicating limited sensitivity to the total variation regularization within the tested range.

Table 6
Ensitivity analysis of major physics-guided regularization weights under lane 1 with a 10% CAV penetration rate.

Case	λ_w	λ_{cw}	λ_a	λ_{ct}	λ_{tv}	RMSE (m/s)
Reference setting	0.10	0.18	0.05	0.00	0.008	1.420
λ_w low	0.05	0.18	0.05	0.00	0.008	1.403
λ_w high	0.20	0.18	0.05	0.00	0.008	1.794
λ_{cw} low	0.10	0.09	0.05	0.00	0.008	1.538
λ_{cw} high	0.10	0.36	0.05	0.00	0.008	1.522
λ_a low	0.10	0.18	0.025	0.00	0.008	1.473
λ_a high	0.10	0.18	0.10	0.00	0.008	1.498
λ_{tv} low	0.10	0.18	0.05	0.00	0.004	1.423
λ_{tv} high	0.10	0.18	0.05	0.00	0.016	1.460
Continuity on	0.10	0.18	0.05	0.02	0.008	1.534

In contrast, the wave-propagation weight λ_w has a stronger influence on reconstruction performance. A smaller value of $\lambda_w = 0.05$ slightly improves RMSE from 1.420 to 1.403 m/s, whereas an overly large value of $\lambda_w = 0.20$ increases RMSE to 1.794 m/s. This suggests that the wave-propagation constraint is beneficial when applied with a moderate strength, but an excessively large weight may over-constrain the reconstructed traffic states and reduce numerical accuracy. In addition, activating the weak conservation-based continuity term with $\lambda_{ct} = 0.02$ increases RMSE to 1.534 m/s, which is consistent with the ablation results and supports setting λ_{ct} to zero in the final model.

Overall, the sensitivity analysis indicates that the proposed framework is reasonably robust to moderate changes in most regularization weights, while the wave-propagation and continuity-related terms should be carefully controlled. These results further support the final hyperparameter setting adopted in this study. More importantly, they help explain the trade-off between physics-guided regularization and point-wise reconstruction error: moderate physical constraints can improve the structural plausibility and propagation consistency of the reconstructed traffic field, whereas overly strong constraints may suppress local variations and lead to increased numerical errors.

5. Conclusion

This study investigated traffic state reconstruction under heterogeneous partial observations in mixed traffic flow, where sparse mobile observations from CAVs and localized fixed observations from RSUs coexist. To address the challenges arising from source heterogeneity, localized continuous missingness, and insufficient recovery of propagation structures, a source-aware physics-enhanced grid fusion framework was developed for lane-level time–space traffic state reconstruction.

Specifically, the proposed method first projects heterogeneous CAV and RSU observations onto a unified regular time–space grid and constructs a multi-channel representation that jointly encodes observed values, observability masks, and explicit spatiotemporal coordinates. On this basis, a block-wise masking self-supervised training strategy is introduced to better match realistic continuous missing patterns in traffic sensing. Furthermore, a source-aware grid fusion network is designed to explicitly distinguish and adaptively integrate mobile and fixed sensing sources through separate branches and gated fusion. To improve the structural plausibility of the reconstructed traffic field, physics-guided regularization terms, including state-aware wave propagation, congestion continuity, and anisotropic smoothing constraints, are incorporated to encourage more realistic congestion evolution and directional organization in the time–space plane.

Experimental results on the NGSIM US-101 dataset demonstrate the effectiveness of the proposed framework. Compared with representative baselines, the proposed method achieves stronger reconstruction performance under self-supervised settings without relying on dense ground truth. The results further show that increasing the CAV penetration rate consistently improves reconstruction accuracy, although the performance gain gradually exhibits diminishing marginal returns at higher penetration levels. In addition, lane-level analysis reveals clear heterogeneity in reconstruction difficulty across lanes, indicating that observation coverage, traffic-state complexity, and propagation characteristics jointly influence model performance. Visual comparisons further confirm that the proposed model is more capable of preserving the continuity, directional consistency, and structural integrity of low-speed congestion bands, thereby better capturing the formation and propagation of traffic waves.

Overall, this study provides a unified solution for multi-source traffic state reconstruction in mixed traffic environments by integrating heterogeneous sensing fusion, self-supervised contextual recovery, and traffic-flow-inspired physical guidance. The proposed framework not only improves numerical reconstruction accuracy, but also enhances the recovery of physically meaningful spatiotemporal propagation structures.

Several directions deserve further investigation in future work. First, the current framework focuses on lane-level reconstruction and can be extended toward joint multi-lane or corridor-network modeling to better capture lane-changing interactions and cross-lane propagation effects. Second, the current physics-guided regularization is introduced in a weakly constrained manner, and tighter coupling with richer traffic flow models may further improve physical consistency. Third, the present study is evaluated on a single benchmark dataset with predefined RSU layouts and CAV penetration settings; broader validation under more diverse traffic scenarios and sensing configurations is still needed. In addition, real-world mixed traffic sensing data may contain measurement noise, localization errors, detection errors, communication delays, asynchronous sampling, and temporary sensor failures. These

imperfections were not systematically examined in the current study due to the lack of synchronized real-world CAV-RSU datasets with reliable ground-truth traffic states. Future work will further evaluate the robustness of the proposed framework under noisy and imperfect sensor conditions when such data become available. It should also be noted that the RSU coverage layout was predefined rather than optimized in this study, and the sensitivity of reconstruction performance to different RSU placement strategies was not systematically examined. Since the same RSU configuration was used across all compared methods and CAV penetration scenarios, the current experiments mainly evaluate model performance under a controlled roadside sensing setting. Future work will further investigate different RSU deployment strategies and their influence on reconstruction accuracy. Finally, beyond reconstruction, the proposed framework may be further extended toward downstream tasks such as traffic state prediction, anomaly detection, and proactive traffic control under mixed traffic conditions.

CRediT authorship contribution statement

Jing Gan: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Conceptualization. **Dapeng Zhang:** Writing – review & editing, Writing – original draft, Formal analysis. **Wei Bai:** Writing – review & editing, Writing – original draft, Formal analysis. **Jiacheng Yin:** Writing – review & editing, Conceptualization. **Linheng Li:** Writing – review & editing, Resources, Conceptualization. **Bin Ran:** Supervision, Resources, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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